

8:44 am; 26th March, 2026

Inside premises: No visible activity

Outside premises: Wet soil, vegetation, promising

10:40 am

Figuring out how their monitor works.

Have set up cameras, haven't started recording.

19:05

Horrible morning. I'm not the most technologically challenged, at times I'm surprisingly rather adept with it. But today morning, I came close to admitting defeat, bereft of words to express my astonishment, I still haven't the faintest of ideas why some equipment that worked yesterday didn't work today. The only explanation I can fathom is because I used the same battery plugs as input for the 'small screen' which is really an antique reverse cam for a car in the 80s. Oh well, that at least forced me to explore the grounds a bit more & lo and behold! I found *H. scabiosae* :)

Beautiful, shiny, ChatGPT told me that it's a hoverfly. Gemini recanted with ardour maintaining that it's H. scabiosae. ChatGPT failed. Having found enough nests for 4 cameras (another one gave way), I was satisfied after all. My thesis didn't sink, the ship's still afloat, desperately needing repair, the water's out for now though. After activity dwindled into oblivion around 3 pm, I ordered a heavy meal at a quaint establishment at Skala Katoni and gawked at Balkan terrapins till my phone near died. I went to check on them bees around 18:00 and was rudely driven out by pestilent hordes of swarming mosquitoes. Now seated on my huge bed with clean sheets, with the air conditioner relegated to a heater and a dinner buffet awaiting me, I feel as close as to a white, privileged researcher from a

first-world country conducting extremely tedious & gruelling fieldwork in whatever's left of the wilderness in a Mediterranean second world country. The only starking difference lies in the fact that I still don't know if I'm paying for this out of my pocket or not. I really hope the personal travel grant greets me affirmatively in the morrow. Worried and a bit lonesome, Amey

8:59 am, 27th March

First bees came out.

Definitely correlated with light intensity.

Hover-bombing bee flies.

Peeping bees.

Doesn't look like scabiosaes though. These ones are super small.

9:23 am.

Clouds above us; no activity.

The nests stand out like bullet holes. Each species marking the earth every which way, unique and architecturally incomprehensible.

★ Record wind velocity as well.

9:37 am.

No activity, moderate-strong winds.

A mosaic of plants surrounds the nests. I still can't estimate the total # nests because I feel I might count the ant/alien bee nests too.

Whenever there's a moment of sunny respite, I see some activity in the bees.

10:15 am

Started raining, recorded 3 vids ~ 20 min max duration

From different angles.

10:50 am

Stopped raining, getting a bit sunny again.

11:16 am.

Warm again. No signs of bees yet (as I write this a bee swooped for my ear as though reprimanding me.) After a brief shower, the nest holes aren't the most evident. I see a moth fluttering about (strange for the morning right?) or maybe it's not a moth and is one of those countless insects I know so little about. I'm curious to see them emerge, the bees I mean. I suspect the one I spent most of the morning looking at is *Lassioglossum malachurum*. Since they are so abundant, I might do a comparative study.

Dunno time (phone's recording) ~ 15 min. later

I guess it's still a bit cold for *A. scabiosa* to come out. A lot of the smaller insects (ants, potential *L. malachurum* & some even smaller

bees) are out right now when the sun's kisses aren't so wet and floppy. But a lot of the

'bullet holes' are smoothened out now and

it's really difficult to tell if there are nests here in the first place. My phone's recording

Site 3 (completely missed this yesterday eve.)

Just to test out the resolution of my iPad, I'll record the bee colony in front of me.

12:28 pm.

No activity of any bees. Just ants & beetles. Will

take a short lunch break now.

But first, let's do a round about field site 1 & record some plants.

1. *Hordium murinum* (Wall/mouse barley)
H. scabiosae likes Asteraceae (H. stiles, knapweed)
thrives in disturbed habitats.

2. (Spiny southwicket) *Sonchus asper*
Asteraceae

3. *Ranunculus repens* (creeping buttercup)

4. *Hypochaeris radiata* (wild dandelion)

(12:42 pm: see some bees around)

5. *Raphanus raphanistrum* (wild radish)

6. *Trifolium repens* (persian clover) - lots of nests beneath these shrubs - surround the site.

13:09

I suspect there are at least 3 distinct species of bees nesting here: i. H. scabiosae ii. L. malachurum (possibly) iii. a small yellow-bellied bee.

14:13

No bee activity @ site 1.

I think the irradiance is too much right now.

So noone's out spare a few ants and beetles which seem to be constants in any weather.

Leaf-cutter ants labouriously fuss around a nest that seems newly excavated.

14:24

Spotted my 2nd H. scabiosae of the day.

Spotted, another. At least 2 nests @ site 3

14:45

L. malachurum still out. iPad's recording a nest @ site 3; unfortunately not much activity.

So, L. malachurum & H. scabiosae are nesting side by side in a hamlet with disproportionately ↑ L. malachurum visible today than H. scabiosae. Maybe their small size helps them get warmer sooner & thus they have a ↑ plastic thermal tolerance whereas

H. scabiosae, being considerably larger, can't warm up quickly enough to forage around this time of the day, and so it's energetically ↑ favourable to just stay home. It is quite windy though right now but with ample sunlight so I wonder what must be the cue that tells bees about this & since L. malachurum came outside their nests pretty quickly after the rain stopped, does the nest architecture somehow tell the bees that it's not wet outside any more. I'm still not sure about H. scabiosae though, maybe I'll focus just on L. malachurum.

Or maybe it was just too windy today for *H. Grabisae* to fly (should be the opposite I'd have imagined given their ↑ size but maybe it's something to do with their foraging preferences?) & each species has a certain

'wind tolerance' given its unique ecology.

@ 15:06 I called it quits & will now set up 2/3 bee traps to collect them tomorrow morning.

28th March, Saturday

10:34 am

Species to look up: a white butterfly with black dots;
a ~~marked~~ ^{mostly cloudy} Barn Swallow.

Intermittently cloudy + sunny right now. Weak wind.

A-mellifera activity near hives.

A Bumblebee's flitting about (good sign due to incy)

No bee activity at sites 0, 1 & 2.

2-3 *L. malachurum* @ site 3.

Deployed iPad to see at least 2 other species of ground-nesting bees. Shooting @ 1080p, 60fps for 10 min to check video size outside a nest with an ind. with its head guarding the nest.

Today's objectives: i. Catch those 2 species and send 'em to Victor to ID.

ii. Test T & Humidity probes & extract data.

11:22 pm

Lot of activity. *L. malachurum* + another species

(or *L. malachurum* with pollen, have to confirm.

DSH23#F5 deploying ^{Hydro} Humidichromé - started @ 11:40 am.

12:10 pm

It's starting to rain. Cloudy, no sun, low winds.

No visible activity. Except ants.

12:22 pm

It was the briefest of showers. Stopped now.

3 hydrochrons are pelling T & humidity but can't start a new mission.

DSH23#F5 487400; 486EC8;
486D3A

Started 4A65641 DSH22L thermochron

@ 12:27. (records every 60 sec. same as the previous hydrochron)

Kept in #6 box.

Started 4A6EE41 DSH22L thermochron

@ 12:32. (same settings as before)

Kept in #1 box.

CB000000 4A663F41 DSH22L thermochron

started @ 12:36 pm.

Kept in #^A box.

0E000004A(AB94) D52422 thermochron
started @ 12:39; #2 box.

[After doing this, stay in the field till 3 pm;
observe; call Kaloni electronic guys eat in Gkabi Kaloni;
observe terrapins; after returning research Gofers, read book]

878880004A(88F74) D61422L thermochron
started @ 12:42; #7 box.

580000004A(8F2A4) D61422L thermochron
started @ 12:45; #10 box.

[Extract data from 1^{pm} - whenever you
switch off (7 pm maybe) & compare using
R how close they are place 'em in the field
for realistic conditions.

(700000004A(9314) D61422L thermochron
started @ 12:49; #9 box.

730000004A(87114) D61422L thermochron
started @ 12:52; #8 box.

✓ Try hydrochrons 1 last time before definitively
marking them as "non-functional"]

8B0000004A(60D4) D61422L started @
12:55; #3 box.

#5 box is empty, the 3 non-functional
iButtons are kept in the box.

13:13

Some bees are out & about. It's cloudy, low winds.

I see butterflies, bee fly (?), l. male chum (?).

Just saw 1 H. sabiosae today, think I know its nest now.

13:17

Really sunny rn. Saw another H. sabiosae @ site 2.

A cricket is nearby, I'm curious what its response
would be. Okay I just saw L'm ind. try to enter
and got kicked out. I've got to record this.

H. sabiosae foraging rn on Rabarbarus (buttercup)
@ site 2.

✓ Maybe H.s. usurped its nest?

13:41

Lots of H. sabiosae around right now - foraging.

Theba pisana: common garden snails around.

Quite aggressive foragers.

Bumblebees around (clearly it was a good sign).

The mountains are clad in shadowy musins,
seductive as the clouds part, their bare interior
reflects a million different rays.

They seem to love persian clovers.

There are 60 many inds rn; I haven't

the faintest where they emerged from.

13:54

There's so much activity rn; I literally had
to leave my iPad recording some random
flowers as a proxy of H. sabiosae activity.

& literally ran away. I'll go back in half an hour (in 14:30) to check again. They really are noisy & seem aggressive. They picked the right day to come out. Also, with so many foundresses, I think I can at least get 10-15 inds for Cmax experiments.

11:42

Moderate-high winds, chilly, cloudy, no visible bee activity. Found their nest. Calling it quits for today.

29th March, Sunday

12:16

In the spirit of Sundays, I start quite late today.

No visible bee activity. I'm worried about a water pipe that lets out water every once in a while near here. Cold, cloudy.

12:29

Sunny m. *L. malachurum*'s out & about.

Lots of *L. malachurum* returning after gathering pollen.

I need to record *M. scabris* coming out of their suspected nest. Lot of activity at site 1 & 2.

H. scabris are coming out at site 2.

Some inds. Flit about at site 3.
(H. sub?)

12:39

It has started to rain a bit. I'll wait for a

bit otherwise shall start transferring data from iButtons. Maybe they nest at diff. locs this time of the yr & then switch nests? Like m. it's at this base of clover & buttercup plants (protected from rain) but later in the season, they might opt for ↑ open nests more exposed to sunlight where soil isn't affected by nearby plants? Or maybe there are multiple entrances to the same nest & they just take the less exposed nest entrance to enter rn when it's raining. shall come back when it's less gloomy & ↑ sunny.

13:03

Raining quite heavily rn. Number 2 malfunctioned;

shall test it again when I get back to Mytilene.

What concerns me is that the clocks need to be resynchronized each time which can screw everything. Some clocks are only a few seconds off, some by an hour!

Enabled #10 at 13:21, shall check its data on 5th April.

13:49

Hygrochron data failed. No date/time recorded! No clue why. Shall switch it on again. Okay, date/time weren't recorded in any of these. Shall start a new mission in 4 by enabling this feature; otherwise shall infer it by ^{recorded} start time. Hygrochron started @ 14:08. #9 started @ 14:09

14:29

Really hungry so shall eat first. Then retrieve iPad, video call my parents. Take a book & go for a walk outside. Have dinner at 19:30. Pack stuff, talk to Michelle about transport tomorrow.

I'm staying here in for 6 nights, $6 \times 35 = \pounds 175$. This is 10 sampling days.

time. 14-25th April: 10 nights, potentially $\pounds 350$; 2nd week of May: 11th - 17th; 7 sampling days starting from 11th return 18th morning: 7 nights, potentially

$\pounds 245$ - Ideally, 11-20th May leave 21st May, potentially $\pounds 350$.

So, in potentially: $\pounds 175 + 350 + 245 = \pounds 770$ which is still out of budget ($\pounds 650$) but manageable.

Ideally $\pounds 875$ unless Michelle reduces rates a bit.

If she does $\pounds 30$ / night which works out

now $\pounds 175 + \pounds 600$

for me even in the ideal scenario.

15:39

I'm pleasantly surprised to find several bees out right now. Have yet to ID the bee I stored in a vial still dunno if it's L.m or some other species. My dad told me to get

smth precautionary for bee stings which I agree. As I do seem to run the risk of clashing into a foraging bee. Maybe, I should record 6 hrs from 10 am - 4 pm.

17:17

Everything's so dear now. DST started, so bees are still out. And it's so sunny even right now. The few ^{past} gloomy days made me forget how warm it actually gets towards summer.

30/03

At uni in Mytilene now. Checking out weather data from meteo.aegean.gr. Agia Paraskevi-Kalloni station is my nearest weather station. Offers hourly measurements for T, humidity, precipitation, wind speed, wind direction, solar radiation, Atm. pressure.

Aegean islands: Lesbos, Limnos, Chios, Rhodes, Gyros

★ Find out how correlated meteo. aegean.gr data is to your T & H data from Burton.

i. check humidity.

ii. check T.

Variable info:

Air T: in $^{\circ}\text{C}$

Cumulative precipitation per hour: in mm.

Relative air humidity: %

Atm. pressure: in hectopascals / millibars

Fuel moisture content: %.

Directⁿ: N, E, W, S: Current speed & gusts (kmph)
Hydrological yr: Oct - ^{current}Sept^{next}

RAWS: Remote Automatic Weather Station.

Fuel moisture: how much H_2O is contained in
vegetatⁿ / organic material that could burn.

Rain-TOT: Total accumulated rain fall

RainDur60: Rain duratⁿ over 60 min: # min.
during which rain was recorded.

Temp-Fuel-AVG: Avg. Fuel moisture

↓
grass, leaves, small twigs, forest litter.

RH-Fuel-AVG: Avg. Fuel relative humidity.

↓
avg. relative humidity in immediate
env. of Fuel (vegetatⁿ) over the hr.

Avg. air T typically recorded ~ 2m. Above ground.

WS-S-WVT: Wind speed Scalar - Vector Average
↓
takes both mag. & dir.
into consideration.

WD-D1-WVT: Wind Directⁿ - Vector Average.

0°: North

90°: East

180°: South

270°: West

Patm-AVG: Average Atmospheric Pressure

1 hPa = 1 mb

Typically @ sea level: ~ 1013 hPa

Imp. vars. for me:

Rain-TOT, RainDur60, T-soil-AVG, T-soil-MAX,

T-soil-MIN, Tair-AVG, Tair-MAX, Tair-MIN,

RH-AVG, RH-MAX, RH-MIN, WS-S-WVT, WS-MAX,

WD-D1-WVT, Patm-AVG, Patm-MAX, Patm-MIN,

Net-Rad-AVG : 18 Variables!

↓
Avg. Net Radiatⁿ

(incoming - outgoing)

W/m².

Next: see if iButtons corroborate with each other
& with meteo. agency gr. data.

Before this:

I think I saw a *Systropha* bees

↓
oligotectic → exclusively visit Convulvul
flowers.

Viktor:

In social species, (T_{max} may not change seasonally
but (T_{min} will be subjected to Δs during
spring/summer.

L. malachurum forages throughout the day in summer

W. Mediterranean → both occur.

Halictidae → very early.

Cardiidae

+ Chironium

+ mites → wo mites, subitose
brambles - Ciratim

jellederalcz@gmail.com

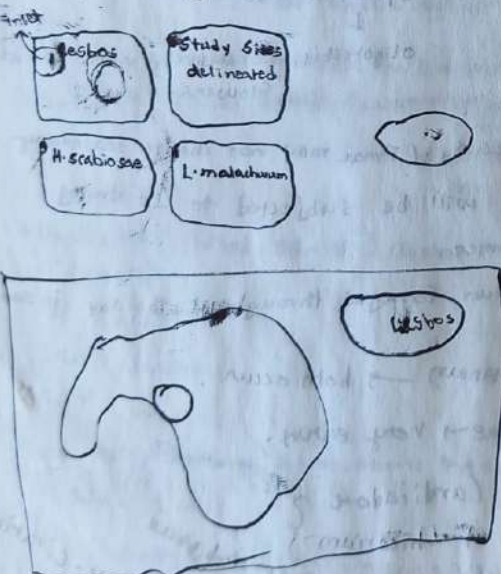
Okay, so Jelle gave me a lot of pointers to think about:

- i. Competitive exclusion with *H. sexcinerus* & *L. malachurum*.
- ii. Both species occur in W. Mediterranean
- iii. *H. scabiosae* prefer *Chicorium* & *Cardioides* (sub-genera from *Asteraceae*).
- iv. doesn't have milk grooves; *Sexcinerus* has 'em

2nd April, Thursday

Started mock trials with Field batteries for GoPro & Sony Action camera @ 10:40.

Study site Figure:



Pau Senna

+34 646 04 7625

Gabrielle's contact

Good Thursday & Friday; noon walk around in churches

Friday night: Seafood & ouzo at night

★ No significant diff. for *L. malachurum*!

14/4/26

Hydrochion just recorded 4 days -

mission start time:

Apr 14 15:42:01 - EEST 2026

Number 9 started @ 15:46

recorded for 7 days!

Number 1 started @ 15:50 → just recorded a day!

Number 2 started @ 15:54:01 same: 1

Number 3 started @ 15:57:01

Number 8 started @ 16:02:01

Number 4 started @ 16:04:01

Number 6 started @ 16:06:01

Number 10 started @ 16:06:01

Number 7 started @ 16:10:01

9 thermochrons. 1: Ambient - 2 at surface - 1 at

15/04/26, 9:40 am.

Taking my first sample soon. A bit nervous. Shall jot down light intensities after 10 min.

Current plan:

10-10:10

11-11:10

12-12:10

13-13:10

14-14:10

15-15:10

16-16:10

17-17:10

if 10-10:10 doesn't work out, 15-15:10

Or maybe wait & see, if you don't see any bee at 10.

Shift to 10:30

As to yesterday's worst case:

10:30-10:40

11:30-11:40

12:30-12:40

13:30-13:40

14:30-14:40

15:30-15:40

16:30-16:40

17:30-17:40

8 x 10 min vids x

5 cameras

everyday!

$\frac{400 \text{ min}}{5} = 80 \text{ min!} = 6.67 \text{ hrs per day!}$

7 hrs.

11 days + 7 days = $18 \times 7 \text{ hrs} = 126 \text{ hrs}$

can analyse

25 Apr - 11 May: 16 days = $\frac{126}{16} = 7.875 \text{ hrs. everyday!}$

Should've noted zoom for my phone. Anyways now it's at

4.3x. Recording started in 10 am. Stop recording when action camera says 10, because time's last to switch on. Activity started in 10, nests opened before I switched 1st record.

I forgot to note yesterday we found a pygmy grasshopper near the nests. The honeybee activity started around 10 am today & maybe *Sistophis*? Still James has to ID.

I see insect activity near site 2 - Hope 4.3x works on my phone - Victor said the nests (*H. sabineae*) are shallow, ~30 cm deep.

I'm going to sample 10 days: 15-24 April & 7 days: 11-17 May.

11 nights

6 nights

17 x 35

595 €

Field observations: site 1 → 2 pygmy

grasshoppers, *L. malta* 1, *H. scab* 2

↓ displaying atop *H. scab* towers?

Lots of bumblebees & honeybees out in

Maybe I shouldn't stand here?

H. scab flies as though on acid. Central mound →

Common for *L. M* & *H. S* nests.

↓ back (Nwards)
front (face SE)

131613 - site 2

5560200 - site 1

11:44:

A bit cloudy, I see bees ^{crawling} inside grass. Lotta activity near nests. Loads of honeybees around. Barn swallows sing oh-so eloquently.

in 7-8 inds near nest (site 1)

Do revisions today; → Answer reviewer H's sat thanks to read whole thing one last time rev-3, & fuck off to rev. 1.

* You should turn on Thanasis' actⁿ camera 5 min. before so as to not disturb the bees.

I'm pissed with the hotel staff. They really see me as an interloper +1 that they must lose with even though they know I'm a student who can't afford much. Even though I'm not paying what they expect rich Europeans / American tourists to, I'm still shelling out a lot. Given the facilities thus far, I'm more than satisfied though & shouldn't really care how people treat me. It's not worth wasting energy over.

Site 2: run → 10743)

1:14 → 9881168

★ Limitation of study: Svetla's camera shut off during 12:00 - 12:10 & no vid was

recorded: r

Just to be safe, ASD card is Svetla's

camera after 4 runs.

Lotta H.S & L.M activity in.

Personal Obs: ↓ inside a nest, small beetle outside tolerated.

★ Thanasis' camera isn't started! :C

Site 1: 54073

2: 45378

14:03 ↓

Lotta activity. H.S. chases grasshopper away? Check this in iPad video.

Beautiful emerald-turquoise blow fly nearby.

13:03 J

Site 2: 104250

1

Kerami: myloc

19:26

↓ 19:52 → sunset

Today: 13-22°C, Aug: 16°C

10:10 → 17°C

16:35 → 20°C

10:34 → 16°C

18:13 → 19°C

11:16 → 19°C

19:16 → 18°C

12:04 → 20°C

20:12 → 17°C

13:07 → 21°C

21:46 → 16°C

00:00 → 15°C

Humidity:

10:00 → 66%, dew pt (DP): 10°
 10:05 → 65%
 10:15 → 64%
 10:23 → 63%
 10:33 → 62%
 10:43 → 61%
 10:56 → 60%
 11:06 → 59%
 11:19 → 58%
 11:28 → 57%
 11:43 → 56%
 11:55 → 55%
 12:10 → 54%
 12:28 → 53%
 12:51 → 52%
 13:16 → 51%
 13:35 → 50%
 13:49 → 49%
 14:12 → 48%
 14:36 → 47%
 15:19 → 48%
 15:31 → -11-, DP: 11°
 15:35 → 49%
 15:52 → 50%
 16:05 → 51%
 16:27 → 52%
 16:40 → 53%
 16:54 → 54%
 17:11 → 55%
 17:24 → 56%
 17:39 → 57%
 17:54 → 58%
 18:09 → 59%
 Today, avg. humidity: 60%
 DP: 9-12°

Pressure: Avg: 1.018 hPa, best: 1.015 hPa

10:00: 1.019 hPa
 12:59: 1.018 hPa
 14:44 → 1.017 hPa
 17:26 → 1.016 hPa

Total day

1st light: 6:11

Sunrise: 6:38, Sunset: 19:52,

Last light: 20:20, total daylight: 13 hr 13 min

April: 6:38-19:52 Avg. sunrise & sunset

Wind: 1-9 km/h, gusts up to 14 km/h

10:00: 4 NNE, gusts: 10 kmph

10:08: 5 NNE

10:34 → 5 NE; 14 kmph

11:02 → 6 NE

11:39 → 7 NE, 17 kmph

12:34 → -11-, 16 kmph

13:35 → 7 NNE; 17 kmph

14:18 → 8 NNE; 17 kmph

14:41 → 9 N, 19 kmph

15:36 → -11-, 18 kmph

16:40 → 8 N, 18

17:25 → 7 N

17:38 → 7 NNW, 15 kmph

16/4

Day 2 of sampling

10:03 am

No activity as yet. It's a bit more chilly today compared to yesterday. Hopefully it doesn't rain. Even the honeybees aren't out yet. Yesterday I saw an interesting behaviour: grasshoppers displaying atop nest mounds. Swallows & mosquitoes are out. Everything's super quiet on the soil. Not a single insect in sight. Mornings like these, beautiful as they may be replete with bird chirps & a fresh scent of mist-laden mountains, scare me for the beings nestled snug in the soil yet sense it cold and treacherous to venture out.

L47: site 2 → 18540

site 1 → 10211

11:15:-

41250 - site 1

12:15:-

site 2: 30602

site 1: 30950

13:15

44426 → site 2

45466 → site 1

14:15

site 2: 101829

site 1: 12641

15:15

3246000 → site 2

36043 → site 1

14:15

site 2 → 25080

site 1 → 24287

From my recollect's,

1: Ambient T

2: ground level site 1

3: ground level site 2

4: just underground site 1
(no #5)

6: just underground site 2

7: 5 m. } site 1

8: 10 m. }

9: 5 m. } site 2

10: 10 m. }

I think

Observation: yesterday bees moved less as the day progressed, as in feeding on ground rather flowers.

17:13 → site 2 → 11764

site 1 → 12641

relatively
e2
to retain

MI-T: Hydromon T

MI-H: Humidity

Next A: 18/4

20/4

22/4

24/4

★ Note: Sveta's camera issue again. (and full

Field day 1:-

Reflect's: Scared & truly alone, starting out proved a lot guttier than I thought. I started doubting myself & called Johnny to confirm the previous day's activity onset time. Reassured, I started the recordings, with a ball in my stomach, at 10 am. 14th April, to think back, was indeed a marvellous day. Maybe the first camera angles we got were exceptionally good (for Field site 2) & now on my third day, time & want of use made every angle bse their charm. I still think the first day angles were somehow different, somehow in the lazy, hazy sunny glaze, everything shimmered & clarity reigned supreme. After two recordings, shocked & aghast, I saw Sveta's camera died on the third recording & ever since (has just been a day of executing this,

I charge it everytime I end a recording session (after the second session.)

→x→
10^{am} Site 2 → 13395

Site 1 → 15769

11:09:- see some activity

→x→

Site 2 → 28582

Site 1 → 36120 : avg, min & max taken at 11:5)

→x→
Reflect'n for day 1 cont:

All in all, after 2 bee stings & 2 fust-ups (forget to switch or rather check thansis camera), fwas disasters but a learning curve is.

→x→
★ Started 12 pm recording. 3-5 min late, Sveta's rec.

ideal for this.

13:?

Site 2 → 39075

Site 1 → 45546

→x→
13:00

Site 2: 43595

Site 1: 58274

→x→

17:15:

12506 - site 2 -

15752 - site 1

18/4: Day 4 of sampling. 10 am.

Super sunny day today, expect a lot of activity.
The bees are already out. I saw H. sub. in the hotel premises. We found L.M. & H.S. bees in the hotel's premises which makes me quite happy because it means we can collect these inds for thermal limits' exps.

Note: I averted a near catastrophe yesterday when I saw a pipe from the pool been dropped over to this side of the fence & it released H_2O like I hear it is, these sites would've been decimated. I could see my nests vanish before my eyes & pleaded with the gardener to move it. On my second attempt, I was successful in conveying this & now all's good (except maybe the honey bee that's foraging way too close to me rn near the *Raphanum* plant.

I see a fire bug near me.

It's a shame I have to turn off the rec now bec. there's so much activity going on!

Lux: site 2 → 120359

Site 1: 41550 163963

* Mention sampling days, hrs, plants in memos.

10:55 am: Took site 1's lux again because I mistook that reading + seems. There's so much activity rn! I'm

really pleased.

11:00

Spiders, blowflies, H.s. abound @ site 1. I dunno what the spiders & flies do near the resting mounds, maybe sun bather? Or maybe is excavated soil cooler? Well it's darker in color so absorbs heat better or they might just be coming for dropped pollen on the way in?

Note: I saw some new orange flowers near the persian clovers @ site 2.

Chamomile ice-creams: why do their centers bulge so?
Site 1 surrounded by clovers (most relevant) & buttercups & those new orange flowers. It's such a shame to

Beh. scoring:

Watch vid on iPad, score on Boris.

Keys.

→

11:15

Lux site 2 → 101773

Site 1 → 100751

Note: ID what the orange flower is.

4

11:05

I see a wasp nearby. From 13's session, I saw a bee ag near site 1 & a dragonfly!

Lux: Site 2 → 101629

Site 1 → 96690

5:15 J
 Site 2 → 92442
 Site 1 → 12728)

19 Apr: Sun

Weather station website isn't loading.

2 & 4 iButtons (just outside, just beneath soil)

didn't record any data. → site 1

10 am:

133725: site 2

134697: site 1

— x —

11 am:

89325 - site 2

87097 - site 1

— x —

13:00

Site 2 → 86657

Site 1 → 86677

— x —

15:16 I think

Michael just put in the T logger & iButtons
 sensors. And he says Anthophora nearby, have
 to look Hiss up.

Lux → site 2 → 104998
 site 1 → 00000

Bee	Body Weight	Weight	TA	Durat ⁿ	T	Durat ⁿ	T
H.G.	2.866						

Let's standardise cold coma duratⁿ; 20 min seems good.

T_A false OTF;

Add T_i column

8: 40, 25.9

↓
duratⁿ to
right

↓
 T_R

7 → incubated.
1 →

16:437

L. malachurum 7 → transferred to diff. nut mid-trial so adding 1 more min for cold camera & didn't work out.

21st Apr

Day 7, T: 33 am

Waiting for H.S. & L.M. to emerge. Victor asked me yesterday how did I know if these bees started their activity sooner or no. With the season, their activity might be shifting earlier. I did doubt that in six more days there'd be a drastic difference but just to know & record emergence, I'm here. So far it's really damp but the sun is nice & out. I'm eager to do my take-off exps. with H.S., my target is to do at least 5 individuals so that I can do multiple individuals from both species. It feels like I haven't written down in a while & there's so much to tell about the past few days. I was so occupied that I hadn't the least amount of time to jot notes. Firstly, Michael's been incredibly helpful - he gave me 3 cameras! His handheld camera is one of the best cameras I've ever used. The model is Canon Vixia HF5106 with 1920x1080 recording (I still don't know the setting at which I'm recording).

8:30 am:

Thought I saw a bee emerge from site 1 but it

@ 8:15

1

might be just a random insect. I feel there's too much variation in my righting data, so a sample size of more than 10 is ideal. So today I hope I get to sample at least 5 H.S. & see if they follow a similar ethogram of waking up as L.M. I think I have 4 good L.S.

measurements. So need 11 more. For H.S., I start today. To ^{avoid} depleting inds from 1 area, I need to look for H.S. in other areas too. Worst case, my beh. site.

8:40:..

H.S.

Thought I saw L.M. @ site 2 @ 8:40 & definitely L.M. at 8:41 so started my cameras over there. No movement yet @ site 1. Definite emergence @ site 2 @ 8:47. Saw a bee enter site 1 @ 8:48 & come out from diff. site @ 8:49.

24th Apr, Friday, my last day is

This has become my life now, alas it all crescendos to an end. As all good things eventually do, it's coming to an end & it'll be really hard adjusting to life back in Mytilene. I've come to realise too much I like this kind of life, the rush that's persistently swelling, how involved I am in things other than myself. I've come to realise a family in Victor, Michael and Isabella & I'll dearly miss them, most of all Michael. He made me feel heard after so long & has tremendously helped me with equipment. I can't believe today's the last day I spend with his cameras. Oh well. Yesterday was an especially miserable day, I'm still glad I could catch 3 individuals

For my take-off study. I have 10 good samples
for each species & I plan on doing 5 more heavily
for each species but the best would be 20 individuals.
My plan is to sample the other sites as best as I can
today & to consider inds. from my own site for the
final day. Yesterday, I had a funny moment when
essentially I was waiting for a cloud to pass because
bees come out of their ^{on a cold day} nests only when there's a
surge of sunlight that dashes in when the clouds pass
over it & there's a ^{glorious golden} moment that can't be missed.
I was at this H. scab nest where Victor sampled
over the past years & I saw an individual poop out
of its nest & stay there.